# Grade 10 Term 1 Lesson 1 Practice Activity 1
# Decision Alternatives, Events, Consequences, States, and Payoff Tables

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## Evaluated criteria

**C1** Characterises decision-making problems under uncertainty by identifying decision alternatives, states of nature, consequences, and the economic or financial context of the decision.

**C2** Organises decision problems using payoff tables, placing alternatives, states of nature, and payoffs in a clear structure for comparison.

## 1 Introduction

**Problems:**


1.1. Main Ideas and Concepts

1.2. Introductory General Problem (C1–C2)

1.3. Advanced General Problem (C1–C2)


### 1.1 Main Ideas and Concepts

A decision problem under uncertainty describes a choice that must be made before an uncertain event is known. The decision-maker chooses one *alternative*; later, the uncertain event produces one *state of nature*. A *consequence* describes what happens for that alternative–state combination, and a *payoff* is the numerical value assigned to the consequence. Thus, consequences describe outcomes in words, while payoffs express their economic or financial values in numbers.

Let

$$A_1, A_2, \ldots, A_m$$

be the available decision alternatives, where $i$ identifies a particular alternative, and let

$$S_1, S_2, \ldots, S_n$$

be the possible states of nature, where $j$ identifies a particular state. The decision-maker chooses one $A_i$, and the uncertain event subsequently produces one $S_j$.

| Element | Description |
|---|---|
| Alternatives | **A1:** $A_1, A_2, \ldots, A_m$ |
| Statesofnature | **S1:** $S_1, S_2, \ldots, S_n$ |
| Events | The uncertain event that produces one of the states $S_j$ |
| Consequences | The consequence and its payoff $P_{ay}(A_i, S_j)$ for each pair $(A_i, S_j)$ |

The pair $(A_i, S_j)$ produces one consequence, whose numerical payoff is $P_{ay}(A_i, S_j)$. Alternatives form the rows of a payoff table and states form its columns:

| Alternative | $S_1$ | $S_2$ | $\cdots$ | $S_n$ |
|---|---|---|---|---|
| $A_1$ | $P_{ay}(A_1, S_1)$ | $P_{ay}(A_1, S_2)$ | $\cdots$ | $P_{ay}(A_1, S_n)$ |
| $A_2$ | $P_{ay}(A_2, S_1)$ | $P_{ay}(A_2, S_2)$ | $\cdots$ | $P_{ay}(A_2, S_n)$ |
| $\vdots$ | $\vdots$ | $\vdots$ | $\ddots$ | $\vdots$ |
| $A_m$ | $P_{ay}(A_m, S_1)$ | $P_{ay}(A_m, S_2)$ | $\cdots$ | $P_{ay}(A_m, S_n)$ |

In particular, $P_{ay}(A_i, S_j)$ means the payoff obtained when alternative $A_i$ is selected and state $S_j$ occurs.

More advanced problems use the following payoff components.

| Symbol | Meaning |
|---|---|
| $q(A_i, S_j)$ | Number of items for $A_i$ under $S_j$ |
| $v(A_i)$ | Variable payoff per item determined by $A_i$ |
| $v(S_j)$ | Variable payoff per item determined by $S_j$ |
| $f(A_i)$ | Fixed payoff determined by $A_i$ |
| $f(S_j)$ | Fixed payoff determined by $S_j$ |
| $P_{ay}(A_i, S_j)$ | Total payoff for the pair $(A_i, S_j)$ |

The complete four-component structure is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\big[v(A_i) + v(S_j)\big] + f(A_i) + f(S_j).$$

Here, $q(A_i, S_j)v(A_i)$ is the variable payoff generated by the alternative-based rate; $q(A_i, S_j)v(S_j)$ is the variable payoff generated by the state-based rate; $f(A_i)$ is the alternative-based fixed component; and $f(S_j)$ is the state-based fixed component. Adding all applicable components gives $P_{ay}(A_i, S_j)$.

A component whose argument is $A_i$ changes by alternative and therefore by row. A component whose argument is $S_j$ changes by state and therefore by column. A component with both arguments, such as $q(A_i, S_j)$, can change for every cell.

Simpler structures occur when components are absent; an absent component can be treated as zero. For example,

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(A_i)$$

uses only an alternative-based variable component, while

$$P_{ay}(A_i, S_j) = f(A_i) + f(S_j)$$

uses only the two fixed components.

Directly supplied payoffs need no decomposition: when $P_{ay}(A_i, S_j)$ is already given, it is placed directly in its payoff-table cell. The direct-payoff form and the complete four-component form therefore describe the same table structure; they differ only in whether the cell value is supplied or calculated.

A reusable procedure is:

1. Identify the alternatives $A_i$.
2. Identify the possible states $S_j$.
3. Determine the consequence for each pair $(A_i, S_j)$.
4. Calculate or record each payoff $P_{ay}(A_i, S_j)$.
5. Place alternatives in rows and states in columns.

The purpose is to convert a verbal decision situation into a complete, comparable payoff table. At this stage, the purpose is not to select the best alternative.

## 1.2 Introductory General Problem (C1–C2)

A decision-maker chooses between generic alternatives $A_1$ and $A_2$. After the choice, one uncertain event produces either state $S_1$ or state $S_2$. The payoffs, in thousand USD, are supplied directly:

$$P_{ay}(A_1, S_1) = 10, \quad P_{ay}(A_1, S_2) = 4, \quad P_{ay}(A_2, S_1) = 7, \quad P_{ay}(A_2, S_2) = 8.$$

The arguments $A_i$ and $S_j$ identify the alternative (the row) and state (the column), respectively. The four combinations are

$$(A_1, S_1), \quad (A_1, S_2), \quad (A_2, S_1), \quad (A_2, S_2).$$

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | |
|---|---|---|
| Alternatives | **A1:** $A_1$ | **A2:** $A_2$ |
| Statesofnature | **S1:** $S_1$ | **S2:** $S_2$ |
| Events | The uncertain event that produces either $S_1$ or $S_2$ | |
| Consequences | The consequence and its payoff in thousand USD for each pair $(A_i, S_j)$ | |



**C2 - Building the payoff table.**

| Alternative | $S_1$ | $S_2$ |
|---|---|---|
| $A_1$ | 10 | 4 |
| $A_2$ | 7 | 8 |

Because $P_{ay}(A_1, S_1)$ belongs to row $A_1$ and column $S_1$, 10 goes in the first payoff cell. Similarly, $P_{ay}(A_1, S_2) = 4$ goes in row $A_1$, column $S_2$; $P_{ay}(A_2, S_1) = 7$ goes in row $A_2$, column $S_1$; and $P_{ay}(A_2, S_2) = 8$ goes in row $A_2$, column $S_2$.



### 1.3 Advanced General Problem (C1–C2)

A decision-maker chooses $A_1$ or $A_2$ before an uncertain event produces $S_1$ or $S_2$. All monetary values below are in thousand USD, with variable values stated per item:

$$v(A_1) = 2, \quad v(A_2) = 3, \qquad f(A_1) = 5, \quad f(A_2) = 6,$$

$$v(S_1) = 1, \quad v(S_2) = 2, \qquad f(S_1) = 1, \quad f(S_2) = 2.$$

The alternative–state quantities are:

| Alternative | $S_1$ | $S_2$ |
|---|---|---|
| $A_1$ | $q(A_1, S_1) = 2$ | $q(A_1, S_2) = 3$ |
| $A_2$ | $q(A_2, S_1) = 4$ | $q(A_2, S_2) = 5$ |

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | |
|---|---|---|
| Alternatives | **A1:** $A_1$ | **A2:** $A_2$ |
| Statesofnature | **S1:** $S_1$ | **S2:** $S_2$ |
| Events | The uncertain event that produces either $S_1$ or $S_2$ | |
| Consequences | The total payoff in thousand USD for each pair $(A_i, S_j)$ | |



**C2 - Building the payoff table.**

Every cell uses the complete formula

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\,(v(A_i) + v(S_j)) + f(A_i) + f(S_j)$$

The components whose argument is $A_i$ change by alternative and therefore by row. The components whose argument is $S_j$ change by state and therefore by column. The quantity $q(A_i, S_j)$ depends on both arguments and can change in every cell.

| Payoff | Complete numerical substitution | Result |
|---|---|---|
| $P_{ay}(A_1, S_1)$ | $q(A_1, S_1)(v(A_1) + v(S_1)) + f(A_1) + f(S_1) = 2(2+1) + 5 + 1$ | 12 |
| $P_{ay}(A_1, S_2)$ | $q(A_1, S_2)(v(A_1) + v(S_2)) + f(A_1) + f(S_2) = 3(2+2) + 5 + 2$ | 19 |
| $P_{ay}(A_2, S_1)$ | $q(A_2, S_1)(v(A_2) + v(S_1)) + f(A_2) + f(S_1) = 4(3+1) + 6 + 1$ | 23 |
| $P_{ay}(A_2, S_2)$ | $q(A_2, S_2)(v(A_2) + v(S_2)) + f(A_2) + f(S_2) = 5(3+2) + 6 + 2$ | 33 |

Each line displays separately the alternative–state quantity, alternative-based variable value, state-based variable value, alternative-based fixed value, and state-based fixed value before they are combined.

| Alternative | $S_1$ | $S_2$ |
|---|---|---|
| $A_1$ | 12 | 19 |
| $A_2$ | 23 | 33 |



## 2 Direct Payoff Models

**Problems:**


2.1. Local Café Investment (C1–C2)

2.1. Green Energy Production Mix (C1–C2)

2.1. Global Shipping Network Design (C1–C2)

## 2.1 Direct

| **Problems:** |
|---|
| 2.1. Local Café Investment (C1–C2) |
| 2.1. Green Energy Production Mix (C1–C2) |
| 2.1. Global Shipping Network Design (C1–C2) |

**Local Café Investment (C1–C2)**

A small café is deciding whether to launch a new line of artisan desserts. The transaction is daily café sales, where profit is earned from each dessert plate sold. The owner has two options:

1. **Launch the dessert line**, with profit outcomes depending on foot traffic:
   i) High traffic → 42 thousand USD
   ii) Low traffic → 8 thousand USD
2. **Keep the current menu**, with profit outcomes based on historical data:
   i) High traffic → 28 thousand USD
   ii) Low traffic → 20 thousand USD

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | |
|---|---|---|
| Alternatives | **A1:** Launch dessert line | **A2:** Keep current menu |
| Statesofnature | **S1:** High traffic | **S2:** Low traffic |
| Events | Actual foot traffic after the decision | |
| Consequences | Profit in thousand USD for each alternative–state pair | |



**C2 - Building the payoff table.**

| Alternative | High traffic | Low traffic |
|---|---|---|
| Launch dessert line | 42 | 8 |
| Keep menu | 28 | 20 |



**Green Energy Production Mix (C1–C2)**

A renewable-energy company must select an electricity generation mix. The transaction is annual electricity sales, where profit is earned per megawatt-hour generated and sold.

Three weather states are possible:

1. Windy
2. Sunny
3. Cloudy

The company is comparing three strategies:

1. Build wind turbines
2. Build solar farms
3. Build a balanced hybrid system

Modeled profits in thousand USD:

1. Wind turbines → 90 (Windy), 45 (Sunny), 30 (Cloudy)
2. Solar farms → 35 (Windy), 95 (Sunny), 25 (Cloudy)
3. Hybrid system → 70 (Windy), 65 (Sunny), 60 (Cloudy)

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | | |
|---|---|---|---|
| Alternatives | **A1:** Wind turbines | **A2:** Solar farms | **A3:** Hybrid system |
| Statesofnature | **S1:** Windy | **S2:** Sunny | **S3:** Cloudy |
| Events | Weather outcome affecting generation | | |
| Consequences | Profit in thousand USD for each strategy–weather combination | | |



**C2 - Building the payoff table.**

| Strategy | Windy | Sunny | Cloudy |
|---|---|---|---|
| Wind turbines | 90 | 45 | 30 |
| Solar farms | 35 | 95 | 25 |
| Hybrid system | 70 | 65 | 60 |



**Global Shipping Network Design (C1–C2)**

A multinational logistics firm is choosing between two shipping network designs for long-term freight contracts. The transaction is global freight services, where profit is earned per contract delivered on schedule.

Management is considering:

1. **Centralized mega-hub network**
2. **Regional multi-hub network**

The firm models four trade conditions:

1. Trade boom
2. Stable trade
3. Moderate disruptions
4. Severe disruptions

Modeled profits in thousand USD are:

1. Centralized network $\rightarrow 220, 140, 40, -30$
2. Regional network $\rightarrow 180, 160, 110, 60$

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description |
|---|---|
| Alternatives | **A1:** Centralized mega-hub **A2:** Regional multi-hub |
| Statesofnature | **S1:** Boom **S2:** Stable **S3:** Moderate disruptions **S4:** Severe disruptions |
| Events | Trade conditions realized during the planning horizon |
| Consequences | Profit in thousand USD for each network–state pair |



**C2 - Building the payoff table.**

| Network | Boom | Stable | Moderate disruptions | Severe disruptions |
|---|---|---|---|---|
| Centralized mega-hub | 220 | 140 | 40 | -30 |
| Regional multi-hub | 180 | 160 | 110 | 60 |



# 3 One-Component Models

**Problems:**


3.1. Community Bakery Lunch Boxes (C1–C2)

3.1. Rural Internet Subscriptions (C1–C2)

3.2. Coffee Export Routes (C1–C2)

3.2. Lake Ferry Tourism (C1–C2)

3.3. School Meal Service Contracts (C1–C2)

3.3. Mobile Clinic Service Contracts (C1–C2)

3.4. Neighborhood Shelter Support (C1–C2)

3.4. Community Hall Emergency Rentals (C1–C2)


## 3.1 Alternative-Based Variable

**Problems:**


3.1. Community Bakery Lunch Boxes (C1–C2)

3.1. Rural Internet Subscriptions (C1–C2)


### Community Bakery Lunch Boxes (C1–C2)

A community bakery is bidding on office lunch-box contracts. The transaction is meal box sales, where profit is earned per lunch box delivered.

The bakery can choose one pricing strategy:

1. **Standard packaging** with a profit of 0.04 thousand USD per lunch box
2. **Premium packaging** with a profit of 0.07 thousand USD per lunch box

Modeled number of lunch boxes:

1. Standard packaging → 900 (High), 550 (Low)
2. Premium packaging → 650 (High), 250 (Low)

Demand states:

1. High office demand
2. Low office demand

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | |
|---|---|---|
| Alternatives | **A1:** Standard packaging | **A2:** Premium packaging |
| Statesofnature | **S1:** High demand | **S2:** Low demand |
| Events | Actual number of lunch-box contracts realized | |
| Consequences | Profit in thousand USD, based on boxes × profit per box | |



**C2 - Building the payoff table.** Calculation table in thousand USD:

$$\text{Payoff} = \text{number of lunch boxes} \times \text{profit per lunch box}$$

| Alternative | High demand | Low demand |
|---|---|---|
| Standard packaging | $900 \times 0.04 = 36$ | $550 \times 0.04 = 22$ |
| Premium packaging | $650 \times 0.07 = 45.5$ | $250 \times 0.07 = 17.5$ |

| Alternative | High demand | Low demand |
|---|---|---|
| Standard packaging | 36 | 22 |
| Premium packaging | 45.5 | 17.5 |



**Rural Internet Subscriptions (C1–C2)**

A rural internet provider is selecting a subscription plan mix. The transaction is monthly broadband subscriptions, where profit is earned per subscription sold.

Decision alternatives:

1. **Basic plan** with profit of 0.03 thousand USD per subscription
2. **Plus plan** with profit of 0.05 thousand USD per subscription
3. **Pro plan** with profit of 0.08 thousand USD per subscription

Modeled subscriptions sold:

1. Basic → 1400 (High), 900 (Low)
2. Plus → 1000 (High), 700 (Low)
3. Pro → 600 (High), 300 (Low)

Enrollment states:

1. High enrollment
2. Low enrollment

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | | |
|---|---|---|---|
| Alternatives | **A1:** Basic plan | **A2:** Plus plan | **A3:** Pro plan |
| Statesofnature | **S1:** High enrollment | **S2:** Low enrollment | |
| Events | Actual subscription uptake after plan selection | | |
| Consequences | Profit in thousand USD based on subscriptions × profit per subscription | | |



**C2 - Building the payoff table.** Calculation table in thousand USD:

Payoff = subscriptions sold × profit per subscription

| Alternative | High enrollment | Low enrollment |
|---|---|---|
| Basic plan | $1400 \times 0.03 = 42$ | $900 \times 0.03 = 27$ |
| Plus plan | $1000 \times 0.05 = 50$ | $700 \times 0.05 = 35$ |
| Pro plan | $600 \times 0.08 = 48$ | $300 \times 0.08 = 24$ |

| Alternative | High enrollment | Low enrollment |
|---|---|---|
| Basic plan | 42 | 27 |
| Plus plan | 50 | 35 |
| Pro plan | 48 | 24 |



## 3.2 State-Based Variable

**Problems:**

3.2. Coffee Export Routes (C1–C2)
3.2. Lake Ferry Tourism (C1–C2)

**Coffee Export Routes (C1–C2)**

A coffee exporter must choose a shipping route. The transaction is exporting coffee shipments, where profit is earned per shipment delivered.

Decision alternatives:

1. **Route through Port A**
2. **Route through Port B**

Exchange-rate states:

1. Favorable rate
2. Unfavorable rate

Profit per shipment depends on the exchange rate:

1. Favorable → 6 thousand USD per shipment
2. Unfavorable → 2 thousand USD per shipment

Modeled shipments:

1. Port A → 40 (Favorable), 28 (Unfavorable)
2. Port B → 32 (Favorable), 30 (Unfavorable)

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | |
|---|---|---|
| Alternatives | **A1:** Port A route | **A2:** Port B route |
| Statesofnature | **S1:** Favorable rate | **S2:** Unfavorable rate |
| Events | Exchange-rate level that affects profit per shipment | |
| Consequences | Profit in thousand USD based on shipments × profit per shipment | |



**C2 - Building the payoff table.** Calculation table in thousand USD:

$$\text{Payoff} = \text{shipments} \times \text{profit per shipment}$$

| Route | Favorable rate | Unfavorable rate |
|---|---|---|
| Port A | $40 \times 6 = 240$ | $28 \times 2 = 56$ |
| Port B | $32 \times 6 = 192$ | $30 \times 2 = 60$ |

| Route | Favorable rate | Unfavorable rate |
|---|---|---|
| Port A | 240 | 56 |
| Port B | 192 | 60 |



**Lake Ferry Tourism (C1–C2)**

A lake tourism operator is choosing boat sizes for sightseeing tours. The transaction is tour ticket sales, where profit is earned per ticket sold.

Alternatives:

1. **Operate small ferries**
2. **Operate large ferries**

Seasonal states:

1. Peak season
2. Normal season
3. Rainy season

Profit per ticket depends on the season:

1. Peak → 0.05 thousand USD
2. Normal → 0.04 thousand USD
3. Rainy → 0.03 thousand USD

Modeled tickets sold:

1. Small ferries → 2200 (Peak), 1600 (Normal), 1000 (Rainy)
2. Large ferries → 2600 (Peak), 1800 (Normal), 800 (Rainy)

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | | |
|---|---|---|---|
| Alternatives | **A1:** Small ferries | **A2:** Large ferries | |
| Statesofnature | **S1:** Peak | **S2:** Normal | **S3:** Rainy |
| Events | Tourism season and ticket price level | | |
| Consequences | Profit in thousand USD based on tickets × profit per ticket | | |

**C2 - Building the payoff table.** Calculation table in thousand USD:

$$\text{Payoff} = \text{tickets sold} \times \text{profit per ticket}$$

| Alternative | Peak | Normal | Rainy |
|---|---|---|---|
| Small ferries | $2200 \times 0.05 = 110$ | $1600 \times 0.04 = 64$ | $1000 \times 0.03 = 30$ |
| Large ferries | $2600 \times 0.05 = 130$ | $1800 \times 0.04 = 72$ | $800 \times 0.03 = 24$ |

| Alternative | Peak | Normal | Rainy |
|---|---|---|---|
| Small ferries | 110 | 64 | 30 |
| Large ferries | 130 | 72 | 24 |



### 3.3 Alternative-Based Fixed

**Problems:**

3.3. School Meal Service Contracts (C1–C2)
3.3. Mobile Clinic Service Contracts (C1–C2)

**School Meal Service Contracts (C1–C2)**

A school cafeteria provider is choosing a meal-service contract for a short academic program. The transaction is a fixed service contract, where the recorded payoff is tied only to the selected contract type.

Decision alternatives with fixed payoff components:

1. **Basic meal contract** → 18 thousand USD
2. **Extended meal contract** → 32 thousand USD

Participation states:

1. Regular participation
2. High participation

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | |
|---|---|---|
| Alternatives | **A1:** Basic meal | **A2:** Extended meal |
| Statesofnature | **S1:** Regular participation | **S2:** High participation |
| Events | Participation level observed after the contract choice | |
| Consequences | Fixed payoff in thousand USD tied only to the selected contract | |



**C2 - Building the payoff table.**

| Alternative | Regular participation | High participation |
|---|---|---|
| Basic meal | 18 | 18 |
| Extended meal | 32 | 32 |

### Mobile Clinic Service Contracts (C1–C2)

A health nonprofit is choosing a service contract for mobile clinics. The transaction is a fixed service contract, where the recorded payoff is tied only to the selected contract type.

Decision alternatives with fixed payoff components:

1. **Basic outreach contract** → 24 thousand USD
2. **Extended outreach contract** → 38 thousand USD
3. **Full regional contract** → 52 thousand USD

Community-need states:

1. Regular need
2. High need

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | | |
|---|---|---|---|
| Alternatives | **A1:** Basic outreach | **A2:** Extended outreach | **A3:** Full regional |
| Statesofnature | **S1:** Regular need | **S2:** High need | |
| Events | Community-need level observed after the contract choice | | |
| Consequences | Fixed payoff in thousand USD tied only to the selected contract | | |

**C2 - Building the payoff table.** Calculation table in thousand USD:

$$\text{Payoff} = \text{contract fixed payoff}$$

| Alternative | Regular need | High need |
|---|---|---|
| Basic outreach | 24 | 24 |
| Extended outreach | 38 | 38 |
| Full regional | 52 | 52 |


| Alternative | Regular need | High need |
|---|---|---|
| Basic outreach | 24 | 24 |
| Extended outreach | 38 | 38 |
| Full regional | 52 | 52 |

## 3.4 State-Based Fixed

**Problems:**

3.4. Neighborhood Shelter Support (C1–C2)
3.4. Community Hall Emergency Rentals (C1–C2)

### Neighborhood Shelter Support (C1–C2)

A municipal office is deciding how to prepare a temporary shelter service for short emergency responses. The transaction is public-service shelter support, where the recorded payoff is a fixed municipal support amount tied only to the emergency state.

Decision alternatives:

1. **Open the school gym**
2. **Open the sports center**

Emergency states with fixed payoff components:

1. Minor emergency → 12 thousand USD
2. Major emergency → 30 thousand USD

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

### C1 - Interpreting decision alternatives, events, consequences, and states.

| Element | Description | |
|---|---|---|
| Alternatives | **A1:** School gym | **A2:** Sports center |
| Statesofnature | **S1:** Minor emergency | **S2:** Major emergency |
| Events | Emergency level realized after the shelter decision | |
| Consequences | Fixed payoff in thousand USD tied only to the emergency state | |

### C2 - Building the payoff table.

| Alternative | Minor emergency | Major emergency |
|---|---|---|
| School gym | 12 | 30 |
| Sports center | 12 | 30 |

**Community Hall Emergency Rentals (C1–C2)**

A city council is choosing how to reserve community halls for emergency use. The transaction is public-service facility rental, where the recorded payoff is a fixed municipal support amount tied only to the emergency state.

Decision alternatives:

1. **Reserve one central hall**
2. **Reserve two neighborhood halls**

Emergency states with fixed payoff components:

1. Mild emergency → 18 thousand USD
2. Serious emergency → 35 thousand USD
3. Major emergency → 50 thousand USD

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | | |
|---|---|---|---|
| Alternatives | **A1:** One central hall | **A2:** Two neighborhood halls | |
| Statesofnature | **S1:** Mild emergency | **S2:** Serious emergency | **S3:** Major emergency |
| Events | Emergency level realized after the reservation decision | | |
| Consequences | Fixed payoff in thousand USD tied only to the emergency state | | |



**C2 - Building the payoff table.** Calculation table in thousand USD:

$$\text{Payoff} = \text{emergency-state fixed payoff}$$

| Alternative | Mild emergency | Serious emergency | Major emergency |
|---|---|---|---|
| One central hall | 18 | 35 | 50 |
| Two neighborhood halls | 18 | 35 | 50 |


| Alternative | Mild emergency | Serious emergency | Major emergency |
|---|---|---|---|
| One central hall | 18 | 35 | 50 |
| Two neighborhood halls | 18 | 35 | 50 |




# 4 Two-Component Models

**Problems:**


4.1. Recycling Collection Contract (C1–C2)
4.1. Museum Ticket Platform (C1–C2)
4.2. Local Printing Contracts (C1–C2)
4.2. Cold-Storage Contracts (C1–C2)
4.3. Urban Landscaping Bids (C1–C2)
4.3. Solar Installation Bids (C1–C2)
4.4. Regional Bus Charters (C1–C2)
4.4. Air Cargo Charters (C1–C2)
4.5. Fruit Export Permits (C1–C2)
4.5. Seafood Export Licenses (C1–C2)
4.6. Warehouse Security Agreements (C1–C2)
4.6. Harbor Inspection Agreements (C1–C2)


## 4.1 Alternative-Based Variable Plus State-Based Variable

**Problems:**


4.1. Recycling Collection Contract (C1–C2)
4.1. Museum Ticket Platform (C1–C2)


**Recycling Collection Contract (C1–C2)**

A municipal contractor chooses a collection fleet for a two-month recycling service contract.

**Decision alternatives**

1. Electric fleet
2. Diesel fleet

**States of nature**

1. High recovery
2. Low recovery

**Alternative-based variable payoff per item** $v(A_i)$

1. Electric fleet: 0.6 thousand USD
2. Diesel fleet: 0.4 thousand USD

**State-based variable payoff per item** $v(S_j)$

1. High recovery: 0.3 thousand USD
2. Low recovery: 0.1 thousand USD

**Modeled number of items** $q(A_i, S_j)$

1. Electric fleet: 80 under High recovery, 55 under Low recovery
2. Diesel fleet: 95 under High recovery, 65 under Low recovery

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | |
|---|---|---|
| Alternatives | **A1:** Electric fleet | **A2:** Diesel fleet |
| Statesofnature | **S1:** High recovery | **S2:** Low recovery |
| Events | The realized state affecting the transaction | |
| Consequences | Payoff in thousand USD for each alternative–state pair | |



**C2 - Building the payoff table.** The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\,(v(A_i) + v(S_j))\,.$$

Detailed calculations (thousand USD):

| Alternative | High recovery | Low recovery |
|---|---|---|
| Electric fleet | $80 \times (0.6 + 0.3) = 72$ | $55 \times (0.6 + 0.1) = 38.5$ |
| Diesel fleet | $95 \times (0.4 + 0.3) = 66.5$ | $65 \times (0.4 + 0.1) = 32.5$ |

The simplified payoff table is

| Alternative | High recovery | Low recovery |
|---|---|---|
| Electric fleet | 72 | 38.5 |
| Diesel fleet | 66.5 | 32.5 |



**Museum Ticket Platform (C1–C2)**

A museum consortium selects a ticket platform; the transaction is batches of one thousand admissions processed.

**Decision alternatives**

1. Mobile platform
2. Kiosk platform
3. Hybrid platform

**States of nature**

1. Peak season

2. Regular season

**Alternative-based variable payoff per item** $v(A_i)$

1. Mobile platform: 0.5 thousand USD
2. Kiosk platform: 0.4 thousand USD
3. Hybrid platform: 0.4 thousand USD

**State-based variable payoff per item** $v(S_j)$

1. Peak season: 0.2 thousand USD
2. Regular season: 0.1 thousand USD

**Modeled number of items** $q(A_i, S_j)$

1. Mobile platform: 70 under Peak season, 45 under Regular season
2. Kiosk platform: 60 under Peak season, 50 under Regular season
3. Hybrid platform: 65 under Peak season, 55 under Regular season

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | | |
|---|---|---|---|
| Alternatives | **A1:** Mobile platform | **A2:** Kiosk platform | **A3:** Hybrid platform |
| Statesofnature | **S1:** Peak season | **S2:** Regular season | |
| Events | The realized state affecting the transaction | | |
| Consequences | Payoff in thousand USD for each alternative–state pair | | |



**C2 - Building the payoff table.** The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\,(v(A_i) + v(S_j))\,.$$

Detailed calculations (thousand USD):

| Alternative | Peak season | Regular season |
|---|---|---|
| Mobile platform | $70 \times (0.5 + 0.2) = 49$ | $45 \times (0.5 + 0.1) = 27$ |
| Kiosk platform | $60 \times (0.4 + 0.2) = 36$ | $50 \times (0.4 + 0.1) = 25$ |
| Hybrid platform | $65 \times (0.4 + 0.2) = 39$ | $55 \times (0.4 + 0.1) = 27.5$ |

The simplified payoff table is

| Alternative | Peak season | Regular season |
|---|---|---|
| Mobile platform | 49 | 27 |
| Kiosk platform | 36 | 25 |
| Hybrid platform | 39 | 27.5 |



## 4.2 Alternative-Based Variable Plus Alternative-Based Fixed

**Problems:**

4.2. Local Printing Contracts (C1–C2)
4.2. Cold-Storage Contracts (C1–C2)

### Local Printing Contracts (C1–C2)

A local printing shop is choosing a contract format for school material packets. The transaction is printing packets, where profit is earned per packet printed plus a fixed profit associated with the chosen printing setup.

Decision alternatives:

1. **Small printer setup** (fixed profit 12 thousand USD, profit 0.05 thousand USD per packet)
2. **Digital printer setup** (fixed profit 20 thousand USD, profit 0.04 thousand USD per packet)

Modeled packets printed:

1. Small printer → 500 (High), 300 (Low)
2. Digital printer → 700 (High), 400 (Low)

Demand states:

1. High packet demand
2. Low packet demand

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | |
|---|---|---|
| Alternatives | **A1:** Small printer | **A2:** Digital printer |
| Statesofnature | **S1:** High demand | **S2:** Low demand |
| Events | Packet demand realized after the printing setup decision | |
| Consequences | Profit in thousand USD = packets × profit per packet + fixed profit | |



**C2 - Building the payoff table.** Calculation table in thousand USD:

$$\text{Payoff} = \text{packets printed} \times \text{profit per packet} + \text{setup fixed profit}$$

| Alternative | High demand | Low demand |
|---|---|---|
| Small printer | $500 \times 0.05 + 12 = 37$ | $300 \times 0.05 + 12 = 27$ |
| Digital printer | $700 \times 0.04 + 20 = 48$ | $400 \times 0.04 + 20 = 36$ |

| Alternative | High demand | Low demand |
|---|---|---|
| Small printer | 37 | 27 |
| Digital printer | 48 | 36 |



**Cold-Storage Contracts (C1–C2)**

A logistics company is leasing cold-storage space for agricultural pallets. The transaction is storage contracts, where profit is earned per pallet stored plus a fixed profit associated with the chosen facility.

Decision alternatives:

1. **Small facility** (fixed profit 40 thousand USD, profit 0.18 thousand USD per pallet)
2. **Medium facility** (fixed profit 25 thousand USD, profit 0.12 thousand USD per pallet)
3. **Large facility** (fixed profit 10 thousand USD, profit 0.08 thousand USD per pallet)

Demand states:

1. High storage demand
2. Low storage demand

Modeled pallets stored:

1. Small → 300 (High), 220 (Low)
2. Medium → 600 (High), 420 (Low)
3. Large → 900 (High), 700 (Low)

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | | |
|---|---|---|---|
| Alternatives | **A1:** Small facility | **A2:** Medium facility | **A3:** Large facility |
| Statesofnature | **S1:** High demand | **S2:** Low demand | |
| Events | Storage demand realized after the lease decision | | |
| Consequences | Profit in thousand USD = pallets × profit per pallet + fixed profit | | |



**C2 - Building the payoff table.** Calculation table in thousand USD:

$$\text{Payoff} = \text{pallets stored} \times \text{profit per pallet} + \text{facility fixed profit}$$

| Alternative | High demand | Low demand |
|---|---|---|
| Small facility | $300 \times 0.18 + 40 = 94$ | $220 \times 0.18 + 40 = 79.6$ |
| Medium facility | $600 \times 0.12 + 25 = 97$ | $420 \times 0.12 + 25 = 75.4$ |
| Large facility | $900 \times 0.08 + 10 = 82$ | $700 \times 0.08 + 10 = 66$ |

| Alternative | High demand | Low demand |
|---|---|---|
| Small facility | 94 | 79.6 |
| Medium facility | 97 | 75.4 |
| Large facility | 82 | 66 |



## 4.3 Alternative-Based Variable Plus State-Based Fixed

**Problems:**

4.3. Urban Landscaping Bids (C1–C2)
4.3. Solar Installation Bids (C1–C2)

**Urban Landscaping Bids (C1–C2)**

A landscaping company is bidding on city maintenance projects. The transaction is completing landscaping projects, where profit is earned per project plus a fixed profit adjustment from grant conditions.

Decision alternatives:

1. **Residential team focus** (profit 0.6 thousand USD per project)
2. **Commercial team focus** (profit 1.0 thousand USD per project)

Grant states with fixed profit components:

1. Grant available → +10 thousand USD
2. No grant → −5 thousand USD

Modeled projects completed:

1. Residential → 40 (Grant available), 25 (No grant)
2. Commercial → 30 (Grant available), 20 (No grant)

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | |
|---|---|---|
| Alternatives | **A1:** Residential | **A2:** Commercial |
| Statesofnature | **S1:** Grant available | **S2:** No grant |
| Events | Grant outcome affecting the fixed profit adjustment | |
| Consequences | Profit in thousand USD = projects × profit per project + fixed profit | |



**C2 - Building the payoff table.** Calculation table in thousand USD:

$$\text{Payoff} = \text{projects completed} \times \text{profit per project} + \text{grant fixed profit}$$

| Alternative | Grant available | No grant |
|---|---|---|
| Residential focus | $40 \times 0.6 + 10 = 34$ | $25 \times 0.6 - 5 = 10$ |
| Commercial focus | $30 \times 1.0 + 10 = 40$ | $20 \times 1.0 - 5 = 15$ |

| Alternative | Grant available | No grant |
|---|---|---|
| Residential focus | 34 | 10 |
| Commercial focus | 40 | 15 |



**Solar Installation Bids (C1–C2)**

A solar company is bidding on installation contracts. The transaction is installing solar systems, where profit is earned per installation plus a fixed profit adjustment from policy conditions.

Decision alternatives:

1. **Residential focus** (profit 0.9 thousand USD per installation)
2. **Commercial focus** (profit 1.6 thousand USD per installation)
3. **Mixed portfolio** (profit 1.2 thousand USD per installation)

Modeled installations:

1. Residential → 80 (Rebate continues), 50 (Rebate ends)
2. Commercial → 55 (Rebate continues), 40 (Rebate ends)
3. Mixed → 70 (Rebate continues), 45 (Rebate ends)

Policy states with fixed profit components:

1. Rebate continues → +20 thousand USD
2. Rebate ends → −10 thousand USD

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | | |
|---|---|---|---|
| Alternatives | **A1:** Residential | **A2:** Commercial | **A3:** Mixed portfolio |
| Statesofnature | **S1:** Rebate continues | **S2:** Rebate ends | |
| Events | Policy outcome affecting the fixed profit adjustment | | |
| Consequences | Profit in thousand USD = installs × profit per install + fixed profit | | |



**C2 - Building the payoff table.** Calculation table in thousand USD:

$$\text{Payoff} = \text{installations} \times \text{profit per installation} + \text{policy fixed profit}$$

| Alternative | Rebate continues | Rebate ends |
|---|---|---|
| Residential focus | $80 \times 0.9 + 20 = 92$ | $50 \times 0.9 - 10 = 35$ |
| Commercial focus | $55 \times 1.6 + 20 = 108$ | $40 \times 1.6 - 10 = 54$ |
| Mixed portfolio | $70 \times 1.2 + 20 = 104$ | $45 \times 1.2 - 10 = 44$ |

| Alternative | Rebate continues | Rebate ends |
|---|---|---|
| Residential focus | 92 | 35 |
| Commercial focus | 108 | 54 |
| Mixed portfolio | 104 | 44 |



## 4.4 State-Based Variable Plus Alternative-Based Fixed

**Problems:**

4.4. Regional Bus Charters (C1–C2)
4.4. Air Cargo Charters (C1–C2)

### Regional Bus Charters (C1–C2)

A regional transport firm is choosing a charter fleet for school trips. The transaction is bus charters, where profit is earned per charter plus a fixed profit component tied to the selected fleet arrangement.

Alternatives:

1. **Use mini-bus fleet** (fixed profit 15 thousand USD)
2. **Use coach fleet** (fixed profit 25 thousand USD)

Profit per charter depends on fuel costs:

1. Regular fuel → 4 thousand USD
2. High fuel → 2 thousand USD

Fuel-cost states:

1. Regular fuel cost
2. High fuel cost

Modeled charters:

1. Mini-bus → 25 (Regular), 20 (High)
2. Coach → 18 (Regular), 16 (High)

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

### C1 - Interpreting decision alternatives, events, consequences, and states.

| Element | Description | |
|---|---|---|
| Alternatives | **A1:** Mini-bus fleet | **A2:** Coach fleet |
| Statesofnature | **S1:** Regular fuel | **S2:** High fuel |
| Events | Fuel costs affecting profit per charter | |
| Consequences | Profit in thousand USD = charters × profit per charter + fixed profit | |



**C2 - Building the payoff table.** Calculation table in thousand USD:

$$\text{Payoff} = \text{charters} \times \text{profit per charter} + \text{fleet fixed profit}$$

| Alternative | Regular fuel | High fuel |
|---|---|---|
| Mini-bus fleet | $25 \times 4 + 15 = 115$ | $20 \times 2 + 15 = 55$ |
| Coach fleet | $18 \times 4 + 25 = 97$ | $16 \times 2 + 25 = 57$ |

| Alternative | Regular fuel | High fuel |
|---|---|---|
| Mini-bus fleet | 115 | 55 |
| Coach fleet | 97 | 57 |



**Air Cargo Charters (C1–C2)**

An air cargo firm is choosing a charter fleet. The transaction is cargo charters, where profit is earned per charter plus a fixed profit component tied to the chosen aircraft lease.

Alternatives:

1. **Lease narrow-body aircraft** (fixed profit 30 thousand USD)
2. **Lease wide-body aircraft** (fixed profit 50 thousand USD)

Fuel-price states:

1. Low fuel cost
2. Medium fuel cost
3. High fuel cost

Profit per charter depends on fuel prices:

1. Low fuel → 8 thousand USD
2. Medium fuel → 5 thousand USD
3. High fuel → 2 thousand USD

Modeled charters:

1. Narrow-body → 20 (Low), 18 (Medium), 15 (High)
2. Wide-body → 14 (Low), 12 (Medium), 9 (High)

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | | |
|---|---|---|---|
| Alternatives | **A1:** Narrow-body lease | **A2:** Wide-body lease | |
| Statesofnature | **S1:** Low fuel | **S2:** Medium fuel | **S3:** High fuel |
| Events | Fuel prices affecting profit per charter | | |
| Consequences | Profit in thousand USD = charters × profit per charter + fixed profit | | |



**C2 - Building the payoff table.** Calculation table in thousand USD:

$$\text{Payoff} = \text{charters} \times \text{profit per charter} + \text{aircraft fixed profit}$$

| Alternative | Low fuel | Medium fuel | High fuel |
|---|---|---|---|
| Narrow-body lease | $20 \times 8 + 30 = 190$ | $18 \times 5 + 30 = 120$ | $15 \times 2 + 30 = 60$ |
| Wide-body lease | $14 \times 8 + 50 = 162$ | $12 \times 5 + 50 = 110$ | $9 \times 2 + 50 = 68$ |

| Alternative | Low fuel | Medium fuel | High fuel |
|---|---|---|---|
| Narrow-body lease | 190 | 120 | 60 |
| Wide-body lease | 162 | 110 | 68 |



## 4.5 State-Based Variable Plus State-Based Fixed

**Problems:**

4.5. Fruit Export Permits (C1–C2)
4.5. Seafood Export Licenses (C1–C2)

### Fruit Export Permits (C1–C2)

A fruit exporter is choosing an export permit focus. The transaction is exporting fruit crates, where profit is earned per crate plus a fixed profit adjustment tied to seasonal conditions.

Alternatives:

1. **Citrus export focus**
2. **Berry export focus**

Seasonal states with fixed profit components:

1. Strong harvest season → +12 thousand USD
2. Weak harvest season → −8 thousand USD

Profit per crate depends on the season:

1. Strong season → 0.4 thousand USD
2. Weak season → 0.2 thousand USD

Modeled crates exported:

1. Citrus → 200 (Strong), 150 (Weak)
2. Berry → 180 (Strong), 120 (Weak)

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | |
|---|---|---|
| Alternatives | **A1:** Citrus focus | **A2:** Berry focus |
| Statesofnature | **S1:** Strong | **S2:** Weak |
| Events | Seasonal conditions affecting prices and fixed profit adjustments | |
| Consequences | Profit in thousand USD = crates × profit per crate + fixed profit | |



**C2 - Building the payoff table.** Calculation table in thousand USD:

$$\text{Payoff} = \text{crates exported} \times \text{profit per crate} + \text{season fixed profit}$$

| Alternative | Strong | Weak |
|---|---|---|
| Citrus focus | $200 \times 0.4 + 12 = 92$ | $150 \times 0.2 - 8 = 22$ |
| Berry focus | $180 \times 0.4 + 12 = 84$ | $120 \times 0.2 - 8 = 16$ |

| Alternative | Strong | Weak |
|---|---|---|
| Citrus focus | 92 | 22 |
| Berry focus | 84 | 16 |



**Seafood Export Licenses (C1–C2)**

A seafood exporter is choosing a license focus. The transaction is exporting seafood crates, where profit is earned per crate plus a fixed profit adjustment tied to seasonal conditions.

Alternatives:

1. **Frozen fillet focus**
2. **Fresh export focus**

Seasonal states with fixed profit components:

1. Strong tourism season → +30 thousand USD
2. Normal season → +10 thousand USD
3. Stormy season → −15 thousand USD

Profit per crate depends on the season:

1. Strong → 0.7 thousand USD
2. Normal → 0.5 thousand USD
3. Stormy → 0.2 thousand USD

Modeled crates exported:

1. Frozen fillet → 120 (Strong), 100 (Normal), 80 (Stormy)
2. Fresh export → 90 (Strong), 110 (Normal), 70 (Stormy)

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | | |
|---|---|---|---|
| Alternatives | **A1:** Frozen fillet focus | **A2:** Fresh export focus | |
| Statesofnature | **S1:** Strong | **S2:** Normal | **S3:** Stormy |
| Events | Seasonal conditions affecting prices and fixed profit adjustments | | |
| Consequences | Profit in thousand USD = crates × profit per crate + fixed profit | | |



**C2 - Building the payoff table.** Calculation table in thousand USD:

$$\text{Payoff} = \text{crates exported} \times \text{profit per crate} + \text{season fixed profit}$$

| Alternative | Strong | Normal | Stormy |
|---|---|---|---|
| Frozen fillet focus | $120 \times 0.7 + 30 = 114$ | $100 \times 0.5 + 10 = 60$ | $80 \times 0.2 - 15 = 1$ |
| Fresh export focus | $90 \times 0.7 + 30 = 93$ | $110 \times 0.5 + 10 = 65$ | $70 \times 0.2 - 15 = -1$ |

| Alternative | Strong | Normal | Stormy |
|---|---|---|---|
| Frozen fillet focus | 114 | 60 | 1 |
| Fresh export focus | 93 | 65 | -1 |



## 4.6 Alternative-Based Fixed Plus State-Based Fixed

**Problems:**

4.6. Warehouse Security Agreements (C1–C2)
4.6. Harbor Inspection Agreements (C1–C2)

### Warehouse Security Agreements (C1–C2)

A warehouse operator is choosing a security agreement for seasonal inventory protection. The transaction is a fixed security agreement, where the recorded payoff combines a fixed component tied to the agreement and a fixed component tied to the demand season.

Decision alternatives with fixed payoff components:

1. **Standard security agreement** → 20 thousand USD
2. **Enhanced security agreement** → 35 thousand USD

Demand-season states with fixed payoff components:

1. Normal season → 6 thousand USD
2. Festival season → 14 thousand USD

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | |
|---|---|---|
| Alternatives | **A1:** Standard security | **A2:** Enhanced security |
| Statesofnature | **S1:** Normal season | **S2:** Festival season |
| Events | Demand-season state realized during the security period | |
| Consequences | Profit in thousand USD = agreement fixed payoff + state fixed payoff | |



**C2 - Building the payoff table.** Calculation table in thousand USD:

$$\text{Payoff} = \text{agreement fixed payoff} + \text{state fixed payoff}$$

| Alternative | Normal season | Festival season |
|---|---|---|
| Standard security | $20 + 6 = 26$ | $20 + 14 = 34$ |
| Enhanced security | $35 + 6 = 41$ | $35 + 14 = 49$ |

| Alternative | Normal season | Festival season |
|---|---|---|
| Standard security | 26 | 34 |
| Enhanced security | 41 | 49 |



**Harbor Inspection Agreements (C1–C2)**

A port authority is choosing an inspection agreement for harbor facilities. The transaction is a fixed inspection agreement, where the recorded payoff combines a fixed component tied to the agreement and a fixed component tied to the harbor condition state.

Decision alternatives with fixed payoff components:

1. **Standard inspection agreement** → 28 thousand USD
2. **Rapid-response inspection agreement** → 42 thousand USD

Harbor condition states with fixed payoff components:

1. Calm season → 12 thousand USD
2. Busy season → 5 thousand USD
3. Storm-repair season → −10 thousand USD

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | | |
|---|---|---|---|
| Alternatives | **A1:** Standard inspection | **A2:** Rapid-response inspection | |
| Statesofnature | **S1:** Calm season | **S2:** Busy season | **S3:** Storm-repair season |
| Events | Harbor condition state realized during the inspection period | | |
| Consequences | Profit in thousand USD = agreement fixed payoff + state fixed payoff | | |



**C2 - Building the payoff table.** Calculation table in thousand USD:

$$\text{Payoff} = \text{agreement fixed payoff} + \text{state fixed payoff}$$

| Alternative | Calm season | Busy season | Storm-repair season |
|---|---|---|---|
| Standard inspection | $28 + 12 = 40$ | $28 + 5 = 33$ | $28 - 10 = 18$ |
| Rapid-response inspection | $42 + 12 = 54$ | $42 + 5 = 47$ | $42 - 10 = 32$ |

| Alternative | Calm season | Busy season | Storm-repair season |
|---|---|---|---|
| Standard inspection | 40 | 33 | 18 |
| Rapid-response inspection | 54 | 47 | 32 |



# 5 Three-Component Models

**Problems:**


5.1. Water Treatment Modules (C1–C2)
5.1. Freight Locker Network (C1–C2)
5.2. District Heating Supply (C1–C2)
5.2. Port Cold-Chain Service (C1–C2)
5.3. Pharmacy Delivery Plan (C1–C2)
5.3. Timber Processing Line (C1–C2)
5.4. Public Library Outreach (C1–C2)
5.4. Grain Terminal Schedule (C1–C2)


## 5.1 Alternative-Based Variable Plus State-Based Variable Plus Alternative-Based Fixed

**Problems:**


5.1. Water Treatment Modules (C1–C2)
5.1. Freight Locker Network (C1–C2)


### Water Treatment Modules (C1–C2)

A utility chooses a portable treatment design for batches of water delivered to remote communities.

**Decision alternatives**

1. Membrane unit
2. Filtration unit

**States of nature**

1. Dry season
2. Wet season

**Alternative-based variable payoff per item** $v(A_i)$

1. Membrane unit: 0.8 thousand USD
2. Filtration unit: 0.6 thousand USD

**State-based variable payoff per item** $v(S_j)$

1. Dry season: 0.3 thousand USD
2. Wet season: 0.1 thousand USD

**Alternative-based fixed payoff** $f(A_i)$

1. Membrane unit: 8 thousand USD
2. Filtration unit: 12 thousand USD

**Modeled number of items** $q(A_i, S_j)$

1. Membrane unit: 50 under Dry season, 35 under Wet season
2. Filtration unit: 60 under Dry season, 45 under Wet season

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | |
|---|---|---|
| Alternatives | **A1:** Membrane unit | **A2:** Filtration unit |
| Statesofnature | **S1:** Dry season | **S2:** Wet season |
| Events | The realized state affecting the transaction | |
| Consequences | Payoff in thousand USD for each alternative–state pair | |

**C2 - Building the payoff table.** The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\,(v(A_i) + v(S_j)) + f(A_i).$$

Detailed calculations (thousand USD):

| Alternative | Dry season | Wet season |
|---|---|---|
| Membrane unit | $50 \times (0.8 + 0.3) + 8 = 63$ | $35 \times (0.8 + 0.1) + 8 = 39.5$ |
| Filtration unit | $60 \times (0.6 + 0.3) + 12 = 66$ | $45 \times (0.6 + 0.1) + 12 = 43.5$ |

The simplified payoff table is

| Alternative | Dry season | Wet season |
|---|---|---|
| Membrane unit | 63 | 39.5 |
| Filtration unit | 66 | 43.5 |

**Freight Locker Network (C1–C2)**

A parcel operator chooses a locker network for shipment batches handled under two demand states.

**Decision alternatives**

1. Metro lockers
2. Suburban lockers
3. Mixed network

**States of nature**

1. Strong demand
2. Steady demand

**Alternative-based variable payoff per item** $v(A_i)$

1. Metro lockers: 0.5 thousand USD
2. Suburban lockers: 0.4 thousand USD
3. Mixed network: 0.5 thousand USD

**State-based variable payoff per item** $v(S_j)$

1. Strong demand: 0.2 thousand USD
2. Steady demand: 0.1 thousand USD

**Alternative-based fixed payoff** $f(A_i)$

1. Metro lockers: 10 thousand USD
2. Suburban lockers: 14 thousand USD
3. Mixed network: 12 thousand USD

**Modeled number of items** $q(A_i, S_j)$

1. Metro lockers: 80 under Strong demand, 55 under Steady demand
2. Suburban lockers: 65 under Strong demand, 60 under Steady demand
3. Mixed network: 75 under Strong demand, 65 under Steady demand

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | | |
|---|---|---|---|
| Alternatives | **A1:** Metro lockers | **A2:** Suburban lockers | **A3:** Mixed network |
| Statesofnature | **S1:** Strong demand | **S2:** Steady demand | |
| Events | The realized state affecting the transaction | | |
| Consequences | Payoff in thousand USD for each alternative–state pair | | |

**C2 - Building the payoff table.** The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\,(v(A_i) + v(S_j)) + f(A_i).$$

Detailed calculations (thousand USD):

| Alternative | Strong demand | Steady demand |
|---|---|---|
| Metro lockers | $80 \times (0.5 + 0.2) + 10 = 66$ | $55 \times (0.5 + 0.1) + 10 = 43$ |
| Suburban lockers | $65 \times (0.4 + 0.2) + 14 = 53$ | $60 \times (0.4 + 0.1) + 14 = 44$ |
| Mixed network | $75 \times (0.5 + 0.2) + 12 = 64.5$ | $65 \times (0.5 + 0.1) + 12 = 51$ |

The simplified payoff table is

| Alternative | Strong demand | Steady demand |
|---|---|---|
| Metro lockers | 66 | 43 |
| Suburban lockers | 53 | 44 |
| Mixed network | 64.5 | 51 |



## 5.2 Alternative-Based Variable Plus State-Based Variable Plus State-Based Fixed

**Problems:**


5.2. District Heating Supply (C1–C2)

5.2. Port Cold-Chain Service (C1–C2)


### District Heating Supply (C1–C2)

A district energy provider selects a heat source for units supplied during the winter.

**Decision alternatives**

1. Biomass heat
2. Heat pumps

**States of nature**

1. Mild winter
2. Cold winter

**Alternative-based variable payoff per item** $v(A_i)$

1. Biomass heat: 0.5 thousand USD
2. Heat pumps: 0.6 thousand USD

**State-based variable payoff per item** $v(S_j)$

1. Mild winter: 0.1 thousand USD
2. Cold winter: 0.3 thousand USD

**State-based fixed payoff** $f(S_j)$

1. Mild winter: 6 thousand USD

2. Cold winter: −4 thousand USD (negative adjustment)

**Modeled number of items** $q(A_i, S_j)$

1. Biomass heat: 55 under Mild winter, 85 under Cold winter
2. Heat pumps: 65 under Mild winter, 95 under Cold winter

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | |
|---|---|---|
| Alternatives | **A1:** Biomass heat | **A2:** Heat pumps |
| Statesofnature | **S1:** Mild winter | **S2:** Cold winter |
| Events | The realized state affecting the transaction | |
| Consequences | Payoff in thousand USD for each alternative–state pair | |



**C2 - Building the payoff table.** The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\,(v(A_i) + v(S_j)) + f(S_j).$$

Detailed calculations (thousand USD):

| Alternative | Mild winter | Cold winter |
|---|---|---|
| Biomass heat | $55 \times (0.5 + 0.1) + 6 = 39$ | $85 \times (0.5 + 0.3) - 4 = 64$ |
| Heat pumps | $65 \times (0.6 + 0.1) + 6 = 51.5$ | $95 \times (0.6 + 0.3) - 4 = 81.5$ |

The simplified payoff table is

| Alternative | Mild winter | Cold winter |
|---|---|---|
| Biomass heat | 39 | 64 |
| Heat pumps | 51.5 | 81.5 |



**Port Cold-Chain Service (C1–C2)**

A port authority chooses a cold-chain service model for container batches under three operating states.

**Decision alternatives**

1. Automated service
2. Crew service

**States of nature**

1. Clear flow
2. Congestion
3. Inspection surge

**Alternative-based variable payoff per item** $v(A_i)$

1. Automated service: 0.7 thousand USD
2. Crew service: 0.5 thousand USD

**State-based variable payoff per item** $v(S_j)$

1. Clear flow: 0.2 thousand USD
2. Congestion: 0.1 thousand USD
3. Inspection surge: 0.3 thousand USD

**State-based fixed payoff** $f(S_j)$

1. Clear flow: 8 thousand USD
2. Congestion: $-3$ thousand USD (negative adjustment)
3. Inspection surge: 5 thousand USD

**Modeled number of items** $q(A_i, S_j)$

1. Automated service: 80 under Clear flow, 60 under Congestion, 45 under Inspection surge
2. Crew service: 70 under Clear flow, 65 under Congestion, 55 under Inspection surge

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | | |
|---|---|---|---|
| Alternatives | **A1:** Automated service | **A2:** Crew service | |
| Statesofnature | **S1:** Clear flow | **S2:** Congestion | **S3:** Inspection surge |
| Events | The realized state affecting the transaction | | |
| Consequences | Payoff in thousand USD for each alternative–state pair | | |

**C2 - Building the payoff table.** The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\,(v(A_i) + v(S_j)) + f(S_j).$$

Detailed calculations (thousand USD):

| Alternative | Clear flow | Congestion | Inspection surge |
|---|---|---|---|
| Automated service | $80 \times (0.7 + 0.2) + 8 = 80$ | $60 \times (0.7 + 0.1) - 3 = 45$ | $45 \times (0.7 + 0.3) + 5 = 50$ |
| Crew service | $70 \times (0.5 + 0.2) + 8 = 57$ | $65 \times (0.5 + 0.1) - 3 = 36$ | $55 \times (0.5 + 0.3) + 5 = 49$ |

The simplified payoff table is

| Alternative | Clear flow | Congestion | Inspection surge |
|---|---|---|---|
| Automated service | 80 | 45 | 50 |
| Crew service | 57 | 36 | 49 |



## 5.3 Alternative-Based Variable Plus Alternative-Based Fixed Plus State-Based Fixed

**Problems:**

5.3. Pharmacy Delivery Plan (C1–C2)
5.3. Timber Processing Line (C1–C2)

### Pharmacy Delivery Plan (C1–C2)

A pharmacy cooperative chooses a delivery plan for prescription batches served across two traffic states.

**Decision alternatives**

1. Cargo bikes
2. Electric vans

**States of nature**

1. Light traffic
2. Heavy traffic

**Alternative-based variable payoff per item** $v(A_i)$

1. Cargo bikes: 0.6 thousand USD
2. Electric vans: 0.8 thousand USD

**Alternative-based fixed payoff** $f(A_i)$

1. Cargo bikes: 9 thousand USD
2. Electric vans: 4 thousand USD

**State-based fixed payoff** $f(S_j)$

1. Light traffic: 6 thousand USD
2. Heavy traffic: $-5$ thousand USD (negative adjustment)

**Modeled number of items** $q(A_i, S_j)$

1. Cargo bikes: 60 under Light traffic, 45 under Heavy traffic
2. Electric vans: 75 under Light traffic, 55 under Heavy traffic

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | |
|---|---|---|
| Alternatives | **A1:** Cargo bikes | **A2:** Electric vans |
| Statesofnature | **S1:** Light traffic | **S2:** Heavy traffic |
| Events | The realized state affecting the transaction | |
| Consequences | Payoff in thousand USD for each alternative–state pair | |



**C2 - Building the payoff table.** The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(A_i) + f(A_i) + f(S_j).$$

Detailed calculations (thousand USD):

| Alternative | Light traffic | Heavy traffic |
|---|---|---|
| Cargo bikes | $60 \times 0.6 + 9 + 6 = 51$ | $45 \times 0.6 + 9 - 5 = 31$ |
| Electric vans | $75 \times 0.8 + 4 + 6 = 70$ | $55 \times 0.8 + 4 - 5 = 43$ |

The simplified payoff table is

| Alternative | Light traffic | Heavy traffic |
|---|---|---|
| Cargo bikes | 51 | 31 |
| Electric vans | 70 | 43 |



**Timber Processing Line (C1–C2)**

A wood cooperative selects a processing line for lumber batches sold under three market conditions.

**Decision alternatives**

1. Precision line
2. Standard line

**States of nature**

1. Firm market
2. Normal market
3. Weak market

**Alternative-based variable payoff per item** $v(A_i)$

1. Precision line: 0.9 thousand USD
2. Standard line: 0.6 thousand USD

**Alternative-based fixed payoff** $f(A_i)$

1. Precision line: 7 thousand USD
2. Standard line: 13 thousand USD

**State-based fixed payoff** $f(S_j)$

1. Firm market: 10 thousand USD
2. Normal market: 2 thousand USD
3. Weak market: $-6$ thousand USD (negative adjustment)

**Modeled number of items** $q(A_i, S_j)$

1. Precision line: 70 under Firm market, 55 under Normal market, 40 under Weak market
2. Standard line: 85 under Firm market, 65 under Normal market, 50 under Weak market

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | | |
|---|---|---|---|
| Alternatives | **A1:** Precision line | **A2:** Standard line | |
| Statesofnature | **S1:** Firm market | **S2:** Normal market | **S3:** Weak market |
| Events | The realized state affecting the transaction | | |
| Consequences | Payoff in thousand USD for each alternative–state pair | | |

**C2 - Building the payoff table.** The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(A_i) + f(A_i) + f(S_j).$$

Detailed calculations (thousand USD):

| Alternative | Firm market | Normal market | Weak market |
|---|---|---|---|
| Precision line | $70 \times 0.9 + 7 + 10 = 80$ | $55 \times 0.9 + 7 + 2 = 58.5$ | $40 \times 0.9 + 7 - 6 = 37$ |
| Standard line | $85 \times 0.6 + 13 + 10 = 74$ | $65 \times 0.6 + 13 + 2 = 54$ | $50 \times 0.6 + 13 - 6 = 37$ |

The simplified payoff table is

| Alternative | Firm market | Normal market | Weak market |
|---|---|---|---|
| Precision line | 80 | 58.5 | 37 |
| Standard line | 74 | 54 | 37 |



## 5.4 State-Based Variable Plus Alternative-Based Fixed Plus State-Based Fixed

**Problems:**

5.4. Public Library Outreach (C1–C2)
5.4. Grain Terminal Schedule (C1–C2)

### Public Library Outreach (C1–C2)

A library system selects an outreach format for participant groups served under two attendance states.

**Decision alternatives**

1. Mobile library
2. Pop-up branches

**States of nature**

1. High attendance
2. Low attendance

**State-based variable payoff per item** $v(S_j)$

1. High attendance: 0.7 thousand USD
2. Low attendance: 0.4 thousand USD

**Alternative-based fixed payoff** $f(A_i)$

1. Mobile library: 8 thousand USD
2. Pop-up branches: 11 thousand USD

**State-based fixed payoff** $f(S_j)$

1. High attendance: 5 thousand USD
2. Low attendance: $-2$ thousand USD (negative adjustment)

**Modeled number of items** $q(A_i, S_j)$

1. Mobile library: 50 under High attendance, 30 under Low attendance

2. Pop-up branches: 45 under High attendance, 35 under Low attendance

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | |
|---|---|---|
| Alternatives | **A1:** Mobile library | **A2:** Pop-up branches |
| Statesofnature | **S1:** High attendance | **S2:** Low attendance |
| Events | The realized state affecting the transaction | |
| Consequences | Payoff in thousand USD for each alternative–state pair | |

**C2 - Building the payoff table.** The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(S_j) + f(A_i) + f(S_j).$$

Detailed calculations (thousand USD):

| Alternative | High attendance | Low attendance |
|---|---|---|
| Mobile library | $50 \times 0.7 + 8 + 5 = 48$ | $30 \times 0.4 + 8 - 2 = 18$ |
| Pop-up branches | $45 \times 0.7 + 11 + 5 = 47.5$ | $35 \times 0.4 + 11 - 2 = 23$ |

The simplified payoff table is

| Alternative | High attendance | Low attendance |
|---|---|---|
| Mobile library | 48 | 18 |
| Pop-up branches | 47.5 | 23 |

**Grain Terminal Schedule (C1–C2)**

A grain terminal selects a loading schedule for cargo lots handled under two export states.

**Decision alternatives**

1. Day schedule
2. Night schedule
3. Split schedule

**States of nature**

1. Open channels
2. Restricted channels

**State-based variable payoff per item** $v(S_j)$

1. Open channels: 0.6 thousand USD
2. Restricted channels: 0.3 thousand USD

**Alternative-based fixed payoff** $f(A_i)$

1. Day schedule: 12 thousand USD
2. Night schedule: 8 thousand USD
3. Split schedule: 10 thousand USD

**State-based fixed payoff** $f(S_j)$

1. Open channels: 7 thousand USD
2. Restricted channels: $-3$ thousand USD (negative adjustment)

**Modeled number of items** $q(A_i, S_j)$

1. Day schedule: 90 under Open channels, 55 under Restricted channels
2. Night schedule: 75 under Open channels, 60 under Restricted channels
3. Split schedule: 85 under Open channels, 65 under Restricted channels

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | | |
|---|---|---|---|
| Alternatives | **A1:** Day schedule | **A2:** Night schedule | **A3:** Split schedule |
| Statesofnature | **S1:** Open channels | **S2:** Restricted channels | |
| Events | The realized state affecting the transaction | | |
| Consequences | Payoff in thousand USD for each alternative–state pair | | |

**C2 - Building the payoff table.** The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(S_j) + f(A_i) + f(S_j).$$

Detailed calculations (thousand USD):

| Alternative | Open channels | Restricted channels |
|---|---|---|
| Day schedule | $90 \times 0.6 + 12 + 7 = 73$ | $55 \times 0.3 + 12 - 3 = 25.5$ |
| Night schedule | $75 \times 0.6 + 8 + 7 = 60$ | $60 \times 0.3 + 8 - 3 = 23$ |
| Split schedule | $85 \times 0.6 + 10 + 7 = 68$ | $65 \times 0.3 + 10 - 3 = 26.5$ |

The simplified payoff table is

| Alternative | Open channels | Restricted channels |
|---|---|---|
| Day schedule | 73 | 25.5 |
| Night schedule | 60 | 23 |
| Split schedule | 68 | 26.5 |



# 6 Complete Four-Component Models

**Problems:**

6.1. Urban Food Hub (C1–C2)
6.1. Regional Rail Cargo (C1–C2)

## 6.1 Alternative-Based Variable Plus State-Based Variable Plus Alternative-Based Fixed Plus State-Based Fixed

This structure keeps all four payoff components explicit: an alternative-based variable payoff per item $v(A_i)$, a state-based variable payoff per item $v(S_j)$, an alternative-based fixed payoff $f(A_i)$, and a state-based fixed payoff $f(S_j)$. The quantity $q(A_i, S_j)$ may vary for every alternative–state pair, so

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\,(v(A_i) + v(S_j)) + f(A_i) + f(S_j).$$

**Problems:**

6.1. Urban Food Hub (C1–C2)
6.1. Regional Rail Cargo (C1–C2)

**Urban Food Hub (C1–C2)**

A city food hub chooses a distribution system for produce crates delivered under two demand states.

**Decision alternatives**

1. Electric sorting
2. Manual sorting

**States of nature**

1. Strong demand
2. Soft demand

**Alternative-based variable payoff per item** $v(A_i)$

1. Electric sorting: 0.6 thousand USD
2. Manual sorting: 0.4 thousand USD

**State-based variable payoff per item** $v(S_j)$

1. Strong demand: 0.2 thousand USD
2. Soft demand: 0.1 thousand USD

**Alternative-based fixed payoff** $f(A_i)$

1. Electric sorting: 9 thousand USD
2. Manual sorting: 14 thousand USD

**State-based fixed payoff** $f(S_j)$

1. Strong demand: 6 thousand USD
2. Soft demand: $-3$ thousand USD (negative adjustment)

**Modeled number of items** $q(A_i, S_j)$

1. Electric sorting: 70 under Strong demand, 45 under Soft demand
2. Manual sorting: 80 under Strong demand, 55 under Soft demand

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | |
|---|---|---|
| Alternatives | **A1:** Electric sorting | **A2:** Manual sorting |
| Statesofnature | **S1:** Strong demand | **S2:** Soft demand |
| Events | The realized state affecting the transaction | |
| Consequences | Payoff in thousand USD for each alternative–state pair | |

**C2 - Building the payoff table.** The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\,(v(A_i) + v(S_j)) + f(A_i) + f(S_j).$$

Detailed calculations (thousand USD):

| Alternative | Strong demand | Soft demand |
|---|---|---|
| Electric sorting | $70 \times (0.6 + 0.2) + 9 + 6 = 71$ | $45 \times (0.6 + 0.1) + 9 - 3 = 37.5$ |
| Manual sorting | $80 \times (0.4 + 0.2) + 14 + 6 = 68$ | $55 \times (0.4 + 0.1) + 14 - 3 = 38.5$ |

The simplified payoff table is

| Alternative | Strong demand | Soft demand |
|---|---|---|
| Electric sorting | 71 | 37.5 |
| Manual sorting | 68 | 38.5 |

**Regional Rail Cargo (C1–C2)**

A rail operator selects a cargo service design for freight lots moved under three network states.

**Decision alternatives**

1. Express service
2. Flexible service

**States of nature**

1. Clear network
2. Busy network
3. Disrupted network

**Alternative-based variable payoff per item** $v(A_i)$

1. Express service: 0.8 thousand USD
2. Flexible service: 0.6 thousand USD

**State-based variable payoff per item** $v(S_j)$

1. Clear network: 0.2 thousand USD
2. Busy network: 0.1 thousand USD
3. Disrupted network: $-0.1$ thousand USD (negative adjustment)

**Alternative-based fixed payoff** $f(A_i)$

1. Express service: 10 thousand USD
2. Flexible service: 15 thousand USD

**State-based fixed payoff** $f(S_j)$

1. Clear network: 5 thousand USD
2. Busy network: 0 thousand USD
3. Disrupted network: $-7$ thousand USD (negative adjustment)

1. Express service: 85 under Clear network, 65 under Busy network, 40 under Disrupted network
2. Flexible service: 75 under Clear network, 70 under Busy network, 50 under Disrupted network

**Modeled number of items** $q(A_i, S_j)$

**Questions/Tasks.**

**C1** Interpreting decision alternatives, events, consequences, and states.

**C2** Building the payoff table.

**C1 - Interpreting decision alternatives, events, consequences, and states.**

| Element | Description | | |
|---|---|---|---|
| Alternatives | **A1:** Express service | **A2:** Flexible service | |
| Statesofnature | **S1:** Clear network | **S2:** Busy network | **S3:** Disrupted network |
| Events | The realized state affecting the transaction | | |
| Consequences | Payoff in thousand USD for each alternative–state pair | | |



**C2 - Building the payoff table.** The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\,(v(A_i) + v(S_j)) + f(A_i) + f(S_j).$$

Detailed calculations (thousand USD):

| Alternative | Clear network | Busy network | Disrupted network |
|---|---|---|---|
| Express service | $85 \times (0.8 + 0.2) + 10 + 5 = 100$ | $65 \times (0.8 + 0.1) + 10 + 0 = 68.5$ | $40 \times (0.8 - 0.1) + 10 - 7 = 31$ |
| Flexible service | $75 \times (0.6 + 0.2) + 15 + 5 = 80$ | $70 \times (0.6 + 0.1) + 15 + 0 = 64$ | $50 \times (0.6 - 0.1) + 15 - 7 = 33$ |

The simplified payoff table is

| Alternative | Clear network | Busy network | Disrupted network |
|---|---|---|---|
| Express service | 100 | 68.5 | 31 |
| Flexible service | 80 | 64 | 33 |

